Mechanistic Interpretability of EEG Foundation Models via Sparse Autoencoders

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Abstract

EEG foundation models achieve state-of-the-art clinical performance, yet the internal computations driving their predictions remain opaque: a barrier to clinical trust. We apply TopK Sparse Autoencoders (SAEs) across three architecturally distinct EEG transformers: SleepFM, REVE, and LaBraM to extract sparse feature dictionaries from their embeddings. By grounding these features in a clinical taxonomy (abnormality, age, sex, and medication), we benchmark monosemanticity and entanglement across architectures. A single hyperparameter procedure, driven by an intrinsic dictionary health audit, transfers robustly across all three architectures. Via concept steering, we introduce a "target vs. off-target" probe area metric to quantify steering selectivity and reveal three operational regimes: selectively steerable, encoded but entangled, and non-encoded. This framework exposes critical representational failures: "wrecking-ball" interventions that collapse concept probes, and clinical entanglements, such as age–pathology confounding, where it is impossible to suppress one concept without corrupting the other. Finally, a spectral decoder maps these interventions back to the amplitude spectrum, translating latent manipulations into physiologically interpretable frequency signatures, such as pathological slow-wave suppression with relative sparing of the α band.

1 Introduction

EEG foundation models pre-trained on large corpora achieve strong performance across a range of tasks: sleep staging, pathology detection, and brain-computer interface decoding [1–5]. Yet the internal computations that produce these predictions remain opaque. Unlike image models, where attention maps already give a coarse visual story, EEG models operate on multichannel time-series whose clinically relevant events: spindles, K-complexes, δ waves, epileptiform discharges, are transient, channel-distributed, and defined by decades of human expert knowledge [6]. Understanding *what* an EEG foundation model has learned, and *where* it has learned it, is both a scientific question and a clinical-trust requirement.

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Mechanistic interpretability [7, 8] addresses this by reverse-engineering the functional roles of individual directions inside a network. The central obstacle is *superposition* [8]: a transformer layer with d dimensions can represent up to $N \gg d$ features by co-activating them sparsely, making individual neurons uninterpretable. Sparse Autoencoders (SAEs) [9–11] have emerged as the dominant tool for recovering these features in language models. A small but growing literature ports the toolkit to medical imaging [12, 13], protein language models [14], and transformers trained on neural signals [15, 16]. Despite this progress, EEG foundation models remain unaddressed; no prior work has bridged the full path from a frozen EEG encoder to sparse feature dictionaries, clinical concept attribution, and spectrum-level mechanistic explanation across architectures.

Contributions. We make the following contributions:

1. **A Cross-Architecture Interpretability Pipeline:** We introduce a unified framework for interpreting EEG transformers that bridges the gap between high-dimensional embeddings and clinical physiology. The pipeline integrates layer-wise TopK SAEs, concept attribution via Testing with Concept Activation Vectors (TCAV), and concept steering, utilizing a single hyperparameter-selection procedure that remains robust across three distinct architectures: SleepFM, REVE, and LaBraM.

   The interactive framework is available at

   `https://ai.braincapture.dk/mechanistic-interpretability/`

2. **The Clinical Semanticity Taxonomy:** We propose a taxonomy to audit encoder representations by partitioning clinical features into three regimes: Separable (monosemantic), Entangled (polysemantic co-activations), and Dead (semantically uninformative/inactive). This serves as a tool to identify latent clinical biases where labels like age or medication co-activate with pathology.
3. **Selectivity via Concept Steering:** We formalize a probe-based selectivity metric; calculating the area between target and off-target "steering curves" to evaluate the fidelity of model interventions. This allows us to distinguish between selective concept removal and "wrecking-ball" interventions, where feature suppression inadvertently collapses the entire embedding space.
4. **Mechanistic Spectral Explanations:** By re-decoding steered activations through a spectral decoder, we transform abstract latent embedding into human interpretable space. This provides mechanistic evidence for model attributions, allowing experts to verify that a pathology prediction is driven by physiologically relevant features.

The Code is available and openly accessible at `https://github.com/BrainCapture/mechanistic-interpretability-for-eeg-foundation-models/tree/v1.1.0`.

2 Background

2.1 EEG Foundation Models

We study three architecturally distinct EEG transformer models spanning two different self-supervised pretraining objectives (multi-modal contrastive learning and masked-token prediction); these are summarized in Table 1. All three encoders are subsequently finetuned end-to-end on the same binary normal/abnormal classification target which is later used as a clinical concept. Performance metrics can be found in Table 2.

2.2 TopK Sparse Autoencoders

A TopK SAE [17] trained on activations $\mathbf{a} \in \mathbb{R}^d$ learns

$$\mathbf{z} = \mathrm{TopK}\left(\mathbf{W}_{\text{enc}}^{\top} \frac{\mathbf{a} - \boldsymbol{\mu}_{\boldsymbol{\ell}}}{\boldsymbol{\sigma}_{\boldsymbol{\ell}}}, k\right), \quad \hat{\mathbf{a}} = \mathbf{W}_{\text{dec}}\, \mathbf{z} + \mathbf{b}_{\text{dec}}, \tag{1}$$

where $\mathbf{z} \in \mathbb{R}^N$ and $\mathbf{W}_{\text{enc}}, \mathbf{W}_{\text{dec}} \in \mathbb{R}^{d \times N}$, where $\mathbf{W}_{\text{dec}}$ has unit-norm columns (decoder directions $\mathbf{w}_i \in \mathbb{R}^d$); $N \triangleq d \cdot E$ and E is the expansion rate. Instead of an L_1 penalty [18], which encourages

but does not guarantee sparsity, the TopK constraint [10] keeps the k largest pre-activations (computed on inputs standardised with the training-split mean μ_ℓ and standard deviation σ_ℓ of layer ℓ), so exactly k features fire per token.

Table 1: **Encoder overview.** Three architecturally distinct EEG transformers spanning different SSL objectives: multi-modal contrastive learning (SleepFM), masked-token reconstruction (REVE), and masked spectrum prediction (LaBraM) are inspected. SleepFM is pretrained on PSG data (EEG, ECG, EMG, and respiratory) with a multi-modal contrastive objective, while REVE and LaBraM are pretrained on EEG alone. All three are subsequently finetuned end-to-end on the same binary normal/abnormal target.

Encoder	Architecture	d	Layers	Token length	SSL objective	Modality	≈Hours
SleepFM [1]	SetTransformer	128	3	1 sec	MMC (InfoNCE) [19]	PSG	585,000
REVE [2]	Transformer	512	22	1 sec	MAE [20]	EEG	60,000
LaBraM [3]	BERT-style	200	12	1 sec	MSP (VQ / BEiT) [21]	EEG	2,500

2.3 TCAV and Concept Attribution

Concept Activation Vectors [22] define a concept C via a linear classifier separating X_C from $X_{\neg C}$ in representation space, yielding a unit-norm direction $\mathbf{v}_C \in \mathbb{R}^d$. We additionally train a linear probe $\mathbf{w}$ on SAE activations $\mathbf{z} \in \mathbb{R}^N$, assigning each example a sensitivity $S_C(\mathbf{x}) = \mathbf{w}^\top \mathbf{z}(\mathbf{x})$. The TCAV score is the fraction of concept examples with positive sensitivity:

$$\mathrm{TCAV}_C = \frac{|\{\mathbf{x} \in X_C : S_C(\mathbf{x}) > 0\}|}{|X_C|}, \tag{2}$$

with significance assessed against $N_{\mathrm{rand}} = 50$ random-label null CAVs.

3 Methodology

The pipeline runs in four stages on a frozen encoder (Figure 1). Stages I–III use established interpretability components (linear probing [23, 24], dictionary learning [18], concept attribution [25]); the novel contribution is Stage IV (concept steering), which uses the dictionary and the spectral decoder to produce mechanistic, spectrum-level explanations of any TCAV attribution. Throughout, "frozen" denotes the encoder held fixed during SAE training and all interpretability stages: each encoder is first finetuned end-to-end on the binary normal/abnormal target, and every downstream analysis then treats that finetuned encoder as a fixed feature extractor.

3.1 Dataset and Preprocessing

We use a clinical 27-lead EEG dataset collected using BrainCapture's BC-1 comprising 3,036 subjects. Recordings were de-identified, and only patients who gave explicit consent for research use were included. The cohort is 57.8% male and includes both pediatric and adult populations. Recordings averaged 33.8 minutes, with 30.2% (917 sessions) identified as abnormal; of these, 76.4% showed epileptiform discharges. EEG signals are preprocessed using an evolved version of [26]. The pipeline applies band-pass (0.5-70Hz) and notch (50Hz) filtering before resampling to the encoder's native frequency: 128 Hz for SleepFM and 200 Hz for REVE and LaBraM. Recordings are segmented into non-overlapping 60-second windows. Subjects are partitioned into 80% train/10% validation/10% test sets via scikit-learn's `GroupShuffleSplit` with random seed 42.

3.2 Stage I: Spectral Decoder

The spectral decoder (SD) is a shallow MLP that maps a frozen encoder token embedding $\mathbf{t} \in \mathbb{R}^d$ to amplitude and phase for each frequency bin:

$$\hat{A}_\nu,\ (\cos\hat{\phi}_\nu, \sin\hat{\phi}_\nu) = \mathrm{SD}(\mathbf{t}), \quad \nu = 1, \ldots, F. \tag{3}$$

$F = 64$ bins, 0–64 Hz, for all encoders (consistent with the 70 Hz low-pass filter applied in preprocessing; Section 3.1). Phase is parameterised as a unit-vector pair to eliminate the 2π discontinuity. SleepFM tokens are channel-averaged; REVE and LaBraM tokens are per-channel-per-second.

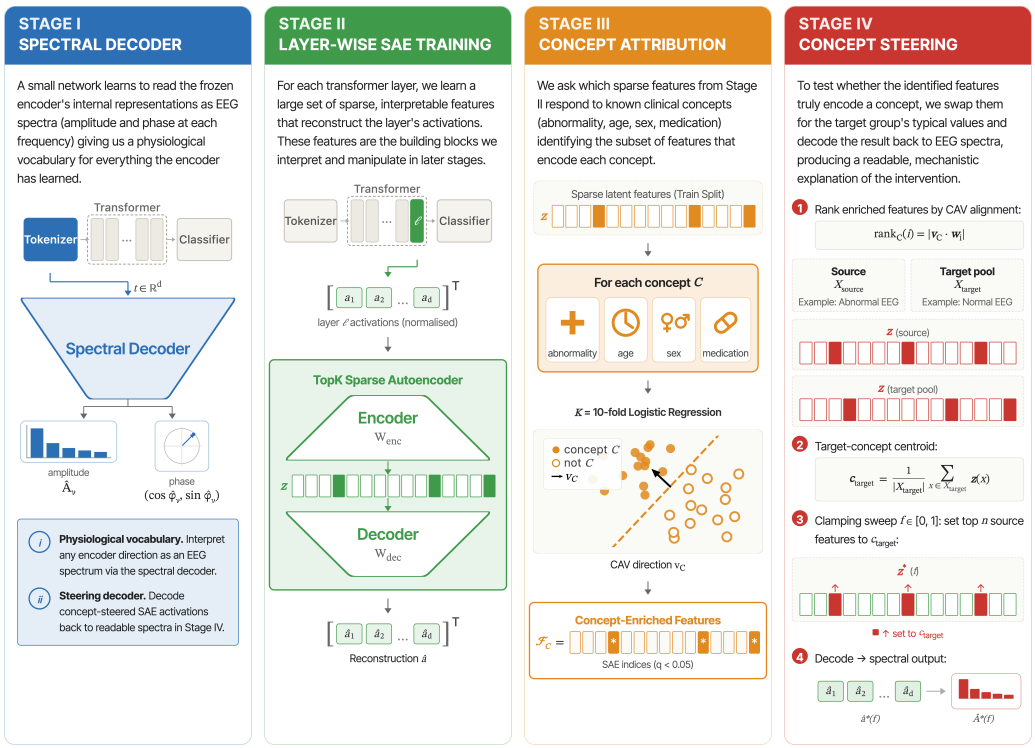


Figure 1: **Pipeline overview.** Starting from a frozen EEG foundation model: (Stage I) A shallow MLP *spectral decoder* translates token embeddings back into a human interpretable space. (Stage II) For each transformer layer, a TopK SAE recovers a sparse, over-complete feature dictionary from normalized encoder activations. (Stage III) SAE features are mapped to known clinical concepts using TCAV. (Stage IV) *Concept steering* substitutes the top-n concept-enriched feature activations with the target-group centroid, decoding the intervention through the SAE and spectral decoder to produce a mechanistically grounded, spectrum-level explanation.

The spectral decoder serves two roles: (i) it provides a physiological vocabulary for interpreting any direction $\mathbf{w} \in \mathbb{R}^d$ via $\mathrm{SD}(\mathbf{w})$, and (ii) it is the final stage of concept steering (Section 3.5).

3.3 Stage II: Layer-wise SAE Training

For each encoder layer ℓ we collect all available training-split token activations, normalise per-dimension, and train a TopK SAE with $E \cdot d$ features. We hold the sparsity budget fixed at $k = 8$ active features per token across the expansion sweep, so that larger dictionaries trade per-feature activity for specialisation at a constant L_0. The spectral worked example (Section 4.5) additionally uses a sparsity-matched SleepFM SAE with $E = 8$, $k = 64$. The reconstruction objective is MSE on normalised activations; dead neurons (never firing in 500 steps) are periodically resampled following [10].

3.4 Stage III: Concept Attribution via TCAV

For each concept $C \in \{$*abnormality, age group, sex, medication (psychiatric), medication (ASM)*$\}$, we obtain $\mathbf{v}_C$ via $K_{\text{fold}} = 10$-fold L_2-regularised logistic regression on dense layer-ℓ activations $\mathbf{a}$, then fit probe $\mathbf{w}$ on $\mathbf{z}$ and compute TCAV_C per Eq. 2. For feature enrichment, we apply a one-sided z-test with Benjamini–Hochberg correction ($q < 0.05$, N simultaneous hypotheses) to identify concept-enriched SAE features, which form the ranking pool for Stage IV.

3.5 Stage IV: Concept Steering

Concept steering provides interventional evidence for the representational fidelity of the SAE dictionary, evaluated against a model-agnostic *selectivity* criterion. Given a target clinical concept C (e.g., age) and a secondary off-target concept (e.g., abnormality), the protocol includes:

1. **Rank by CAV alignment.** The concept direction is mapped onto the sparse dictionary by projecting it onto each SAE decoder direction $\mathbf{w}_i \in \mathbb{R}^d$, giving the alignment-based ranking
$$\mathrm{rank}_C(i) = |\mathbf{v}_C^\top \mathbf{w}_i| \tag{4}$$
2. **Compute the target-concept centroid.** We define the centroid $\mathbf{c}_{\text{target}} \in \mathbb{R}^N$ as the empirical mean of the SAE feature activations across the target dataset X_{target} (e.g., the mean of the "normal" pool when steering abnormal $\rightarrow$ normal):
$$\mathbf{c}_{\text{target}} = \frac{1}{|X_{\text{target}}|} \sum_{\mathbf{x} \in X_{\text{target}}} \mathbf{z}(\mathbf{x}) \tag{5}$$
3. **Clamping features.** For a given intervention fraction $f \in [0, 1]$ and a source sample with SAE activations $\mathbf{z}$, we clamp the activations of the top $n \triangleq \lfloor f \cdot N \rfloor$ TCAV-aligned features to their corresponding scalar values in the target centroid $\mathbf{c}_{\text{target}}$. This produces the intervened sparse activation $\mathbf{z}^*(f)$, which is then decoded back into embedding space:
$$\hat{\mathbf{a}}^*(f) = \mathbf{W}_{\text{dec}}\, \mathbf{z}^*(f) + \mathbf{b}_{\text{dec}} \tag{6}$$

3.6 Evaluating "steerability"

4. **Frozen probe evaluation.** We score the decoded embeddings $\hat{\mathbf{a}}^*(f)$ using two linear probes (target and off-target). Crucially, these probes are fit once on clean ($f = 0$) decoded embeddings and held frozen across the sweep; they act solely as measurement instruments to read out the result of the clamp, completely isolated from the substitution loop.
5. **Excess Selectivity Scalar.** We quantify the quality of the intervention by the integrated area between the off-target and target performance curves:
$$\Delta(\text{rank}) = \int_0^1 \left(\mathrm{AUROC}_{\text{off-target}}(f) - \mathrm{AUROC}_{\text{target}}(f)\right) \mathrm{d}f \tag{7}$$
To account for random feature suppression which may degrade one probe faster than the other, we report the **excess selectivity** $\tilde{\Delta}$ as the difference between the TCAV-ranked area and the expected area under uniformly random feature permutations π:
$$\tilde{\Delta} = \Delta(\text{TCAV}) - \mathbb{E}_\pi[\Delta(\pi)] \tag{8}$$

A value of $\tilde{\Delta} > 0$ indicates a selective push that successfully erases the target concept while sparing off-target axes. Conversely, a value of $\tilde{\Delta} \approx 0$ diagnoses a "wrecking-ball" regime: the intervention is no more selective than random noise, implying that the target concept is so entangled with the global embedding structure that it cannot be independently manipulated. This protocol operationalizes the selectivity criteria used in closed-form concept erasure [27] and amnesic probing [28] for the SAE latent space. Finally, decoding the intervened embedding $\hat{\mathbf{a}}^*(f)$ through the spectral decoder (Section 3.2) yields domain-readable spectral signatures of the intervention (Figure 6).

3.7 Choosing the operating point per model

We identify an optimal layer ℓ^* and SAE expansion ratio E^* for Stages II–IV by sweeping all transformer layers and $E \in \{1, 2, 4, 8, 16, 32, 64\}$ at $k = 8$. We select the configuration that maximizes monosemanticity while penalizing dead features:

$$(\ell^*, E^*) = \arg\max_{\ell, E} \left(\mathrm{separable}_{\ell,E} - \mathrm{dead}_{\ell,E}\right), \tag{9}$$

constrained to layers where target concepts achieve model-level TCAV significance ($p < 0.05$). This taxonomy-driven metric efficiently isolates the model's most transparent regime.

3.8 Taxonomy rule

A feature is *Separable* if it is enriched in exactly one metadata field (age, sex, indication, medication; subject-level Mann–Whitney U, Benjamini–Hochberg-adjusted $p < 0.05$, fold enrichment > 1), *Entangled* if enriched in two or more, and *Dead* otherwise (rarely firing, $< 0.25\,k/N$, and uncharacterised features combined). The p and fold thresholds shift the Separable/Entangled split (Appendix C). The cross-layer interventions (Figures 4–5) use one compact dictionary ($E = 1$, $k = 8$) at every layer, a common operating point across encoders and depths that makes a five-seed sweep tractable.

4 Results

4.1 Binary finetune performance and SAE faithfulness

All three encoders separate the binary normal/abnormal target after end-to-end finetuning. Under 5-fold subject-disjoint cross-validation, a linear probe on mean-pooled embeddings of each finetuned encoder separates normal from abnormal recordings with AUROC 0.94–0.95 (Table 2). As a faithfulness check on the SAE, we replace layer-ℓ activations with their TopK-SAE reconstructions and re-evaluate the same probe on the same test windows: AUROC is preserved within 0.017 at every layer of all three encoders, confirming that the SAE captures the task-relevant structure of the representation rather than discarding it. The effect of interventions on the deployed classifier head itself is evaluated in Section 4.6.

Table 2: **Layer-averaged SAE-faithfulness summary (5-fold CV).** All values are mean $\pm$ std across 5 subject-disjoint folds. Baseline metrics (Balanced Accuracy, F1-Score, and AUROC) are computed for the baseline (no-SAE) linear probe. Mean AUROC (SAE) is the per-layer test AUROC averaged across all transformer blocks of the encoder ($\pm$ is the mean per-fold std averaged across layers, so the unit matches the no-SAE column). Mean Δ is the average AUROC drop from baseline across all layers, with the std taken over layers; Max Δ is the worst-layer drop. The same 5-fold protocol underlies Figure 2.

Encoder	# L	Balanced Acc.	F1-Score	AUROC (no SAE)	AUROC (SAE)	Mean Δ	Max Δ
SleepFM	3	0.901 ± 0.019	0.866 ± 0.047	0.954 ± 0.013	0.951 ± 0.015	0.003 ± 0.002	0.005
REVE	22	0.890 ± 0.018	0.857 ± 0.044	0.944 ± 0.009	0.944 ± 0.009	0.000 ± 0.001	0.006
LaBraM	12	0.875 ± 0.019	0.835 ± 0.045	0.939 ± 0.017	0.933 ± 0.020	0.006 ± 0.006	0.017

The faithfulness check generalises beyond the operating layer. Figure 2 sweeps the SAE-substitution AUROC across every layer of every encoder: SleepFM stays within 0.005 of its no-SAE baseline at all 3 layers, REVE within 0.006 across all 22 layers, and LaBraM within 0.017 across all 12 layers. Interestingly, we see an increase in performance over LaBraM's 12 layers, but from the fourth layer of LaBraM, and for all layers of SleepFM and REVE, the SAE substitution is statistically indistinguishable from the intact encoder regardless of where it is inserted; not just at ℓ^*.

4.2 Spectral decoder reconstruction.

The spectral decoder reliably recovers token-level amplitude spectra from frozen embeddings across all architectures. Overall amplitude R^2 reaches 0.816 ± 0.004 for SleepFM, 0.783 ± 0.004 for REVE, and 0.772 ± 0.004 for LaBraM. Conversely, phase remains unrecoverable across all models (cosine similarity ≤ 0.22), directly reflecting the time-translation invariance inherent to their self-supervised pretraining objectives. All margins represent 95% bootstrap intervals.

4.3 Operating points transfer across encoders

Figure 3 shows the per-encoder operating points (ℓ^*, E^*) selected by the procedure of Section 3.7. Despite differing depth (3/12/22 blocks), width (128/200/512), and training objective, the recipe converges on a common modal expansion: $E^* = 8$ wins at 15/22 REVE, 3/12 LaBraM, and 2/3 SleepFM layers. High separation at low E concentrates at the boundaries: the encoders' first and last transformer blocks collapse to a smaller dictionary (LaBraM L1, L3, L12 and REVE L2, L22 at

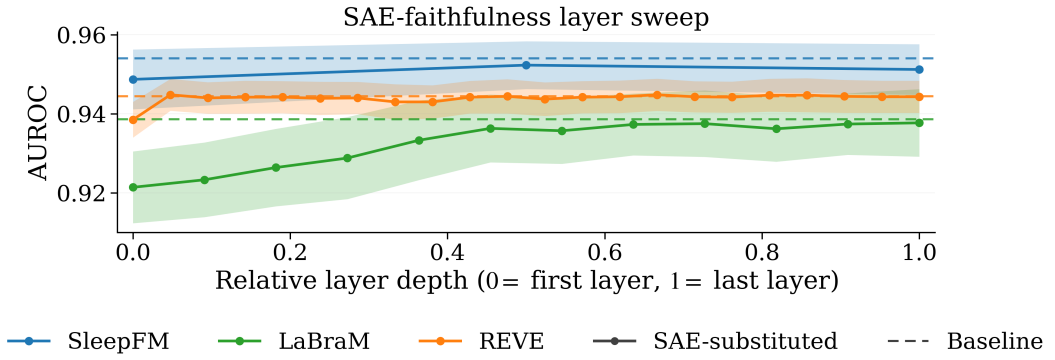


Figure 2: **SAE-faithfulness layer sweep.** Test AUROC of a linear probe trained via 5-fold cross-validation on mean-pooled embeddings of each finetuned encoder. During inference, layer-ℓ activations are replaced by their TopK-SAE reconstructions as ℓ sweeps through every transformer block. Shaded bands represent 95% confidence intervals across the CV folds; the dotted horizontal lines indicate the no-SAE baseline mean. The SAE substitution consistently remains within the baseline performance envelope across all layers and architectures.

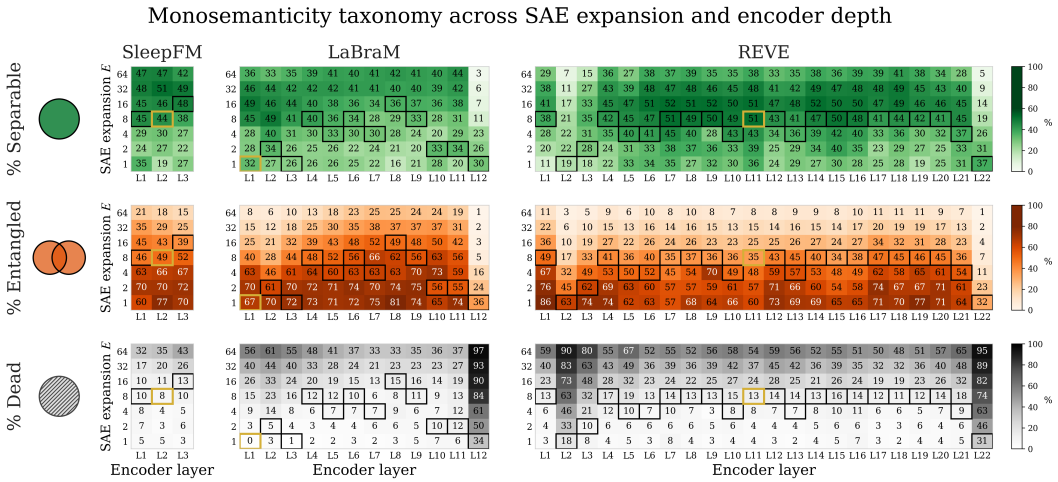


Figure 3: **Monosemanticity taxonomy across SAE expansion and encoder depth.** Each cell reports the fraction of concept-enriched SAE features in one of three taxonomy classes (Separable: monosemantic; Entangled: polysemantic co-activations; Dead: semantically uninformative/inactive). Columns represent encoders (SleepFM, LaBraM, REVE), with x-axes indexing the encoder layer and y-axes indexing expansion factor $E \in \{1, 2, 4, 8, 16, 32, 64\}$. For each layer, we highlight the expansion factor that maximizes the difference between separable and dead feature fractions. The optimum (ℓ^*, E^*) is golden. Find an overview of how E translates to dictionary size in Figure 7.

$E^* = 1$; LaBraM L2, L10, L11 and REVE L3 at $E^* = 2$), consistent with the classification block compressing representations into a low-dimensional task-aligned subspace. SleepFM, with only three transformer blocks, does not compress the representations.

4.4 Cross-model concept selectivity

Abnormality is the finetuning target, so its strong encoding is expected and is not itself the claim; the substantive findings concern concepts absent from the training objective (age, sex, and medication). We apply the substitution-sweep protocol of Section 3.5 to every (encoder, layer, concept) triplet and report excess selectivity $\tilde{\Delta}$ alongside the clean-baseline encoding strength (Figure 4). The two rows together discriminate three regimes:

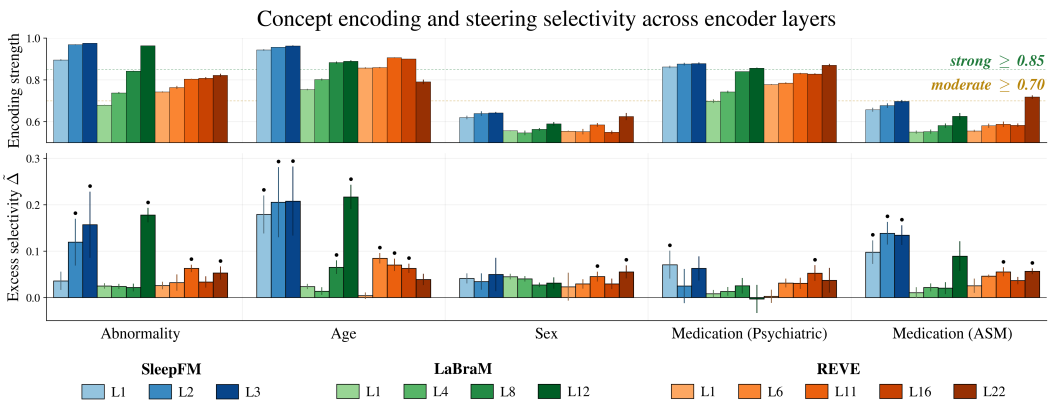


Figure 4: **Concept encoding strength and steering selectivity** ($E = 1$, $k = 8$ at every layer; bars: mean $\pm$ sd over 5 independently seeded SAEs; dots: cells selective in $\geq$ 3/5 seeds, i.e. permutation $p < 0.05$ against 200 random rankings and $\tilde{\Delta} \geq 0.05$). **Top:** Encoding strength (AUROC_0) measured via per-layer linear probes fit to the clean SAE-decoded reconstructions. **Bottom:** Excess selectivity ($\tilde{\Delta}$), quantifying the integrated asymmetry between target and off-target probe degradation under TCAV-ranked clamping (Section 3.5). For *abnormality* as target, we use *age* as off-target. For all other targets, we use *abnormality* as off-target. Together, these metrics map the representational landscape of each model, distinguishing selectively steerable features from highly entangled ones. Exactly how excess selectivity $\tilde{\Delta}$ is computed is illustrated in Figure 5 through examples.

- **Encoded and steerable:** High AUROC_0 + high $\tilde{\Delta}$.
- **Encoded but not steerable:** High AUROC_0 + $\tilde{\Delta} \approx 0$.
- **Weakly encoded:** $\text{AUROC}_0 < 0.7$.

To better illustrate the underlying probe-AUROC sweeps that compute excess selectivity ($\tilde{\Delta}$), in Figure 5, we unpack nine representative cells from the cross-model summary in Figure 4. To understand this grid, the axes should be read as two distinct constraints on the intervention:

Rows define the interventional headroom. The baseline encoding strength dictates the maximum possible impact of the clamp. If a concept is weakly encoded, there is little signal to erase; the target probe already starts close to random guessing, leaving the sweep little room.

Columns map the selectivity regimes. This axis illustrates the continuum from surgical concept removal to unselective collapse. We categorize this into three distinct behaviors:

- **Strong selectivity:** The target probe (red) degrades rapidly under TCAV-ranked clamping while the off-target probe (blue) remains robust. Crucially, the target degrades much faster than the random-clamping baseline, yielding a selectivity area (green) well beyond the random-clamping expectation (grey).
- **Moderate selectivity:** TCAV ranking isolates the concept faster than random noise, but the off-target probe is dragged down alongside it, indicating partial entanglement.
- **Weak selectivity:** The green area shrinks to or below the grey area. Target and off-target representations degrade in lockstep. The intervention removes signal, but it acts as a global "wrecking ball" rather than a targeted push.

Ultimately, this grid provides the qualitative check for the scalar $\tilde{\Delta}$ summaries in Figure 4. It allows the reader to visually verify that a high $\tilde{\Delta}$ reflects an asymmetric, concept-aligned steering effect, rather than a uniform collapse of the layer's representation.

Looking at Figures 4 and 5, examples of encoded and steerable concepts are *age* at every SleepFM layer and at LaBraM L12 ($\tilde{\Delta} = 0.22 \pm 0.03$, 5/5 seeds) and *abnormality* in SleepFM L2–L3 and LaBraM L12 (0.18 ± 0.02). Examples of encoded but non-steerable concepts are *abnormality* and *age* in REVE, encoded at every depth (AUROC_0 0.74–0.91) but steered only weakly ($\tilde{\Delta} \leq 0.09$),

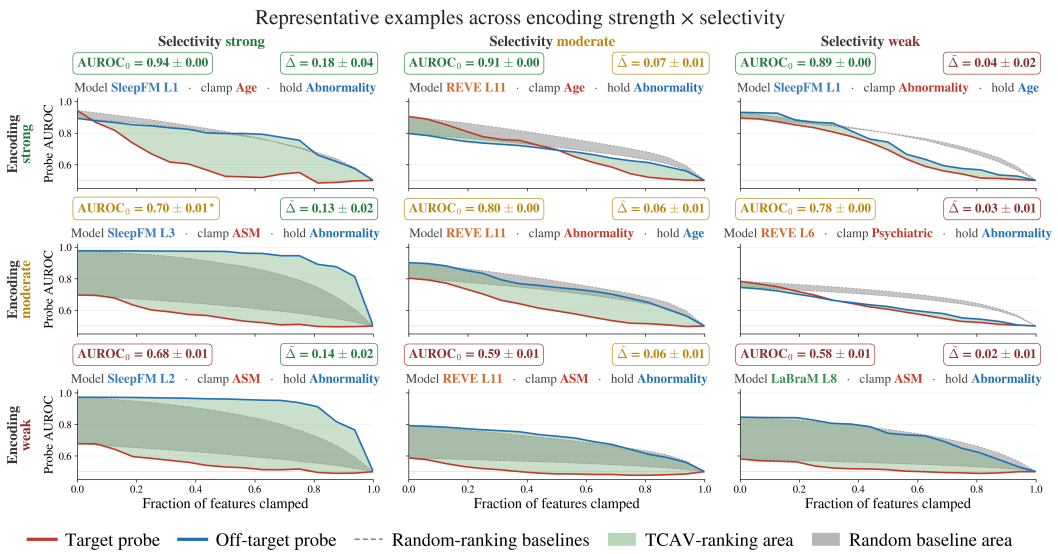


Figure 5: **Steering sweeps across the encoding–selectivity landscape.** Nine representative configurations from Figure 4 arranged by encoding strength (rows) and selectivity $\tilde{\Delta}$ (columns). Panels track target (red) and off-target (blue) AUROC as clamping fraction f increases. Green shading highlights the TCAV-driven selectivity gain while grey shading highlights the random-clamping baseline. Each panel is the cell of Figure 4 closest to its bin centre; no cell falls inside the moderate-encoding, strong-selectivity bin, so that panel shows the cell on its edge (SleepFM L3 ASM; $^\star\text{AUROC}_0 = 0.697$, just below the 0.70 boundary). Curves are means over 5 SAE seeds, and AUROC_0 and $\tilde{\Delta}$ are given as mean $\pm$ sd. The grid contrasts selective steering (left column: the target collapses while the off-target is spared) with the weak-selectivity regime (right column: both probes degrade together, close to the random-clamping baseline). Encoder colours match Figure 4.

and, most clearly, *psychiatric medication* in SleepFM L2 and LaBraM L12 ($\text{AUROC}_0 \geq 0.85$, $\tilde{\Delta} = 0.03 \pm 0.04$ and 0.00 ± 0.03), a "wrecking ball" in which both concepts degrade together. Finally, an example of a weakly encoded concept is sex across all encoders ($\text{AUROC}_0 \leq 0.64$), which functions as the implicit negative control; ASM medication is weakly encoded yet selectively steerable in SleepFM.

4.5 A worked spectrum-level example

Steering a representative abnormal adult EEG toward a normal target corrects canonical pathological markers: clamping the top TCAV-aligned features actively collapses the elevated pathological δ and θ power while restoring the suppressed α peak. By $n = 164$, the intervened spectrum merges into the healthy reference band across the 0.5–45 Hz range. The effect generalises across subjects and encoders (Appendix G): clamping every feature with non-negligible weight in an abnormality probe on adult abnormal recordings lowers δ amplitude by 21.7 ± 1.8 µV in SleepFM ($n = 80$ subjects; negative in $\geq 90\%$ of subjects for every seed) and by 7.9 ± 0.3 µV in LaBraM ($n = 68$; every subject), mean $\pm$ sd over 5 SAE seeds, while α changes far less (-3.8 and -1.0 µV): targeted slow-wave suppression with relative α sparing, not an absolute α increase.

4.6 Baselines, deployed classifier, and public replication

No direction source dominates: PCA and ICA rival the SAE on several concepts, and dedicated erasers (LEACE [27], INLP [29, 28]) remove abnormality with less collateral damage to age (Appendix D); we therefore claim interpretability, not stronger steering. Through the deployed classifier, clamping the top 25% of features lowers SleepFM's abnormality AUROC from 0.91 to 0.76 ± 0.07 (random: 0.88; 10 seeds) on TUAB, less than the probe suggests, so probe-based selectivity is an upper bound (Appendix E). Abnormality steering replicates on TUAB [30, 31] ($\tilde{\Delta} = 0.150 \pm 0.031$, 8 seeds); TUEV [32] events are separable but not selectively steerable (Appendix F).

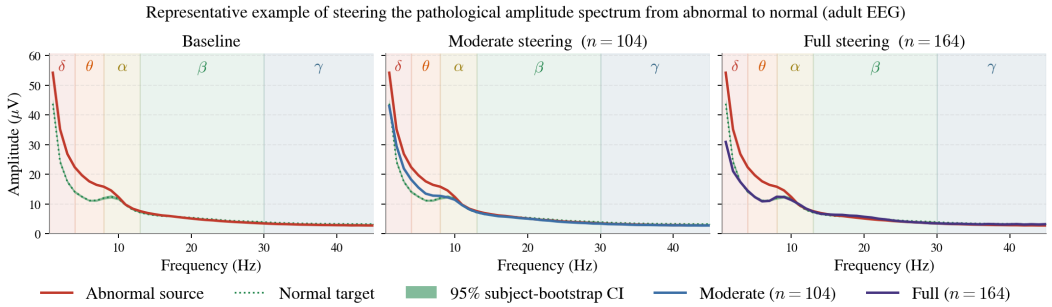


Figure 6: **Spectrum-level concept steering (abnormal → normal).** SleepFM L3 ($E = 8$, $k = 64$); one representative adult recording (population results in Appendix G). Shading denotes 95% bootstrap CIs on the target mean. **Left:** Baseline abnormal source vs. normal target centroid. **Centre & Right:** The decoded spectrum after clamping the top $n = 104$ and $n = 164$ TCAV-aligned features, respectively, to the target centroid.

5 Discussion and limitations

Pretraining versus finetuning. Abnormality is already partially encoded before finetuning: a probe on SAE reconstructions of the pretrained SleepFM reaches AUROC 0.82 on our cohort and 0.73 on TUAB, rising to 0.97 and 0.90 after finetuning. On our cohort, however, abnormality only becomes selectively steerable after finetuning ($\tilde{\Delta} = 0.03 \rightarrow 0.16$), whereas age is encoded and steerable before it ($0.19 \rightarrow 0.22$).

Decoder limitations. Our spectral decoder recovers amplitude but cannot reconstruct phase, as current pretraining objectives discard precise temporal morphology in favor of time-translation invariance. Consequently, while we can steer broad frequency distributions, we cannot generate exact time-domain waveforms. High-fidelity time-domain steering will ultimately require phase-aware foundation models or significantly more expressive decoders.

Steerability vs. selectivity. Any sufficiently aggressive intervention is *steerable*, because destroying all concept information drives a probe to chance. Observational separability does not guarantee selectivity: clamping an observationally pure feature can still trigger a collapse.

Concept taxonomy is investigator-defined. Because our steering protocol (Section 3.5) requires pre-specified concepts, our evaluation is strictly bounded by available cohort metadata. The pipeline verifies rather than discovers. Although inherent to all probe-based methods, this restricts how comprehensively a label-poor dataset can audit a model's latent space.

6 Conclusion

We propose a model-agnostic interpretability framework for auditing the representational health of EEG foundation models. By integrating sparse autoencoders, concept steering, and spectral decoding across three architecturally distinct encoders, the pipeline maps the semantic taxonomy of transformer blocks; isolating where clinical concepts consolidate into selectively steerable features, and where persistent entanglement triggers "wrecking-ball" failures during intervention. By decoding latent representations back into domain-readable spectral signatures, such as focal slowing or alpha-band suppression, we bridge the gap between abstract embedding spaces and clinical physiology, grounding model attributions in mechanistic, frequency-level explanations.

Acknowledgments and Disclosure of Funding

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A Broader impact

This framework is intended as a diagnostic tool to increase transparency and clinical trust in EEG foundation models. By surfacing representational failures such as age–pathology entanglement, it can help practitioners identify and mitigate latent demographic biases. As with all medical AI, frameworks like this should be rigorously validated. The primary risk of misuse is that knowledge of steerable feature directions could inform adversarial manipulation of model predictions; however, this requires white-box access to the encoder, substantially limiting the attack surface.

B Technical appendices and supplementary material

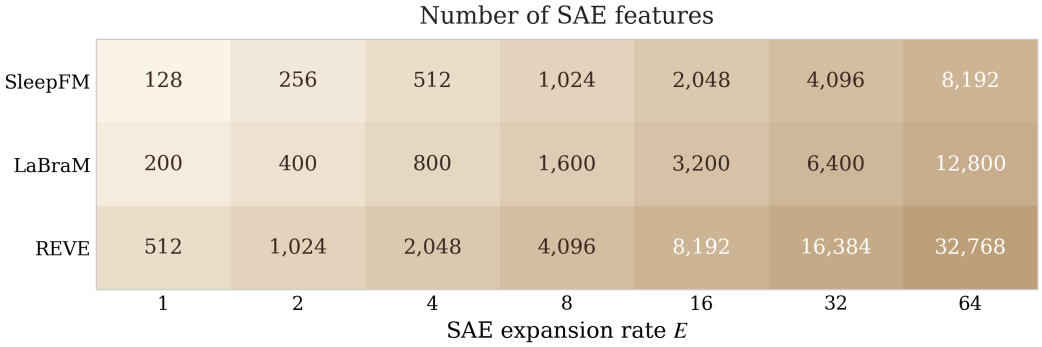


Figure 7: **SAE dictionary size across encoders and expansion rates.** Each cell gives the number of learned SAE features, which equals the encoder's embedding dimension ($d_{\text{enc}} = 128$ for SleepFM, 200 for LaBraM, 512 for REVE) times the expansion rate E. Because REVE is $4\times$ wider than SleepFM, an E=1 REVE SAE already exceeds the size of an E=4 SleepFM SAE, and the E=64 REVE configuration spans 32,768 features. This asymmetry should be kept in mind when comparing taxonomy fractions (Fig. 3) across encoders at matched E: equal E does *not* mean equal capacity.

C Taxonomy rule: threshold sensitivity

Table 6 varies one threshold of the rule in Section 3.8 at a time, holding the others at their defaults (BH-adjusted $p < 0.05$, fold > 1, Dead $< 0.25\,k/N$). The Dead cutoff has no effect over $[0.1, 0.5]\,k/N$. The p threshold moves features between Separable and Entangled, which is why the rule, not a threshold-invariant fraction, is the object we report.

D Baselines, collateral effects, and dedicated erasers

Simpler direction sources. Table 7 replaces the TCAV-ranked SAE features with alternative intervention directions under an otherwise identical protocol. Random sparse dictionaries and centroid steering are no more selective than chance, and dense CAV directions are only weakly selective. PCA is the strongest source for abnormality and ASM medication but is not selective for age or psychiatric medication; ICA and the SAE are selective on all four encoded concepts, the SAE with the larger effect on age and abnormality. No source dominates, so we do not claim that SAE features steer harder; their contribution is that the same intervention is decomposed into individually interpretable features with spectral signatures and a map of what they are entangled with.

Dedicated erasers. LEACE [27] and amnesic probing (INLP [29, 28]) are fit to the concept label itself and are therefore the appropriate tools when the goal is only to remove a concept. At matched suppression (abnormality probe AUROC lowered to ≈ 0.65), they preserve the off-target age probe better than SAE steering (AUROC 0.94 and 0.93 vs. 0.81 ± 0.12 over 10 SAE seeds; PCA 0.92). Steering one concept largely spares the others; the largest collateral effects couple abnormality, age, and psychiatric medication (Figure 8).

Table 3: Notation reference.

Symbol	Type / shape	Definition	First used
Encoder			
d	scalar $\in \mathbb{Z}^+$	Embedding dimension (128 / 200 / 512 for SleepFM / LaBraM / REVE)	Eq. 1
$\mathbf{a}$	vector $\in \mathbb{R}^d$	Layer activation extracted from the frozen encoder	Eq. 1
$\mathbf{t}$	vector $\in \mathbb{R}^d$	Token embedding passed to the spectral decoder	Eq. 3
ℓ	scalar $\in \mathbb{Z}^+$	Transformer layer index	Sec. 3.3
ℓ^*	scalar $\in \mathbb{Z}^+$	Optimal layer selected by Eq. 9	Eq. 9
SAE architecture			
E	scalar $\in \mathbb{Z}^+$	Expansion ratio	Sec. 3.3
$N \triangleq d \cdot E$	scalar $\in \mathbb{Z}^+$	Dictionary size (number of SAE features)	Eq. 1
k	scalar $\in \mathbb{Z}^+$	TopK sparsity budget; $k = 8$ (Fig. 6: $k = 64$ at $E = 8$)	Eq. 1
W_{enc}	matrix $\in \mathbb{R}^{d \times N}$	SAE encoder weight matrix	Eq. 1
W_{dec}	matrix $\in \mathbb{R}^{d \times N}$	SAE decoder; columns $\mathbf{w}_i \in \mathbb{R}^d$ are unit-norm directions	Eq. 1
$\mathbf{w}_i$	vector $\in \mathbb{R}^d$	i-th column of W_{dec}; unit-norm decoder direction	Eq. 1
b_{dec}	vector $\in \mathbb{R}^d$	SAE decoder bias	Eq. 1
$\mu_\ell,\ \sigma_\ell$	vectors $\in \mathbb{R}^d$	Per-dimension mean and std used to normalise activations	Eq. 1
$\mathbf{z}$	vector $\in \mathbb{R}^N$	Sparse SAE feature activation (exactly k non-zero entries)	Eq. 1
$\hat{\mathbf{a}}$	vector $\in \mathbb{R}^d$	SAE reconstruction of the encoder activation	Eq. 1
E^*	scalar $\in \mathbb{Z}^+$	Optimal expansion ratio selected by Eq. 9	Eq. 9
Spectral decoder			
SD	$\mathbb{R}^d \to \mathbb{R}^F \times \mathbb{R}^{2F}$	Shallow MLP mapping a token embedding to amplitude and phase	Eq. 3
F	scalar $\in \mathbb{Z}^+$	Number of frequency bins ($F = 64$)	Eq. 3
$\hat{A}_\nu$	scalar $\in \mathbb{R}$	Predicted amplitude at frequency bin ν	Eq. 3
$\hat{\varphi}_\nu$	scalar $\in [0, 2\pi)$	Phase at bin ν; parameterised as $(\cos\hat{\varphi}_\nu, \sin\hat{\varphi}_\nu)$	Eq. 3
Concept attribution (TCAV)			
$\mathcal{C}$	label	Clinical concept (abnormality, age group, sex, medication)	Sec. 3.4
$\mathbf{v}_C$	vector $\in \mathbb{R}^d$	Concept Activation Vector; unit-norm direction in activation space	Sec. 2.3
$S_C(x)$	scalar $\in \mathbb{R}$	TCAV sensitivity score for example x under concept C	Eq. 2
TCAV_C	scalar $\in [0, 1]$	Fraction of concept examples with $S_C(x) > 0$; above 0.5 $\Rightarrow$ encoded	Eq. 2
N_{rand}	scalar $\in \mathbb{Z}^+$	Number of random-label null CAVs for significance testing; $N_{\text{rand}} = 50$	Sec. 2.3
K_{fold}	scalar $\in \mathbb{Z}^+$	Cross-validation folds for CAV training; $K_{fold} = 10$	Sec. 3.4
Concept steering			
$\text{rank}_C(i)$	function $\mathbb{Z}^+ \to \mathbb{R}$	CAV-alignment score for feature i: $\lvert\mathbf{v}_C \cdot \mathbf{w}_i\rvert$	Eq. 4
X_{target}	dataset	Target pool for the steering intervention	Eq. 5
$\mathbf{c}_{\text{target}} \in \mathbb{R}^N$	vector $\in \mathbb{R}^N$	Target-concept centroid; mean SAE activations over X_{target}	Eq. 5
f	scalar $\in [0, 1]$	Clamping fraction	Eq. 6
$n \triangleq \lfloor f \cdot N \rfloor$	scalar $\in \mathbb{Z}^+$	Number of features clamped (also used bare in Section 4.5 / Fig. 6)	Eq. 6
$\mathbf{z}^*(f)$	vector $\in \mathbb{R}^N$	Intervened activation after clamping top-n features	Eq. 6
$\hat{\mathbf{a}}^*(f)$	vector $\in \mathbb{R}^d$	SAE-decoded embedding of $\mathbf{z}^*(f)$	Eq. 6
$\Delta(\text{rank})$	scalar $\in \mathbb{R}$	Raw selectivity area	Eq. 7
$\tilde{\Delta}$	scalar $\in \mathbb{R}$	Excess selectivity $= \Delta(\text{TCAV}) - \mathbb{E}_\pi[\Delta(\pi)]$	Eq. 8
Hyperparameter selection			
$\text{separable}_{\ell,E}$	scalar $\in [0, 1]$	Fraction of features classified as separable at layer ℓ, expansion E	Eq. 9
$\text{dead}_{\ell,E}$	scalar $\in [0, 1]$	Fraction of features classified as dead at layer ℓ, expansion E	Eq. 9

E Deployed classifier head

Deployed classifier. Propagated through the finetuned classifier heads, clamping the top 25% of TCAV-ranked features lowers the deployed head's abnormality AUROC from 0.91 to 0.76 ± 0.07 for SleepFM finetuned on TUAB (random clamping: 0.88 ± 0.02) and from 0.97 to 0.81 ± 0.11 for LaBraM at its last layer (random: 0.96; 10 SAE seeds each). On the same windows the probe falls further (0.65 and 0.50): the intervention moves the deployed model, but less than the probe suggests, so probe-based selectivity should be read as an upper bound on the deployed effect (Figure 9).

Table 4: Encoder architectures and binary fine-tuning hyperparameters. All three encoders are finetuned end-to-end on the BrainCapture labels using AdamW with two-phase training: a head-only warm-up phase followed by full unfreezing. Learning rate is dropped when the encoder is unfrozen and the weight decay is increased.

	SleepFM	REVE-Base	LaBraM-Base
Where can I find the encoder?	Github	Hugging Face	braindecode and Github
Architecture			
Backbone	SetTransformer	ViT	NeuralTransformer
Embed. dim	128	512	200
Transformer layers	3	22	12
Attention heads	8	8	10
Head dim	16	64	20
MLP ratio	16	2.66 (GeGLU)	4.0
Patch size (samples)	128	200	200
Tokens per window	60	19×66 (overlapping)	19×60
Token receptive field	1.00 s	1.00 s	1.00 s
Token stride	1.00 s	0.90 s	1.00 s
Input data			
Dataset	BrainCapture	Binary (normal/abnormal)	Binary (normal/abnormal)
Sample rate (Hz)	128	200	200
Window length	60	60	60
Channels	27 (10-20)	27 (10-20)	19 (10-20)
Finetuning			
Pooling	temporal_pooling	mean over $C \cdot S$ tokens	mean over patch tokens
Head	`Linear(128,1)`	`Linear(512,1)`	`Linear(200,1)`
Loss	BCE	BCE	BCE
Optimisation			
Optimiser	AdamW	AdamW	AdamW
Total epochs	12	10	15
Head-only w/u ep.	2	2	3
Batch size	32	8	8
LR (head phase)	10^{-3}	10^{-3}	10^{-3}
LR (full phase)	10^{-4}	10^{-4}	10^{-4}
WD (head/full)	0.01 / 0.05	0.01 / 0.05	0.01 / 0.05
Grad-norm clip	1.0	1.0	1.0

Table 5: Estimated compute resources used to produce every stage of the pipeline. All experiments ran on a single Google Cloud VM with one NVIDIA L4 GPU (24 GB VRAM), an Intel Xeon CPU at 2.20 GHz, and 128 GB of system RAM.

Stage	**Per run**	**Wall-clock**	**Peak GPU mem**
SleepFM fine-tune	12 ep, batch 32	$\approx$ 7 min	$\approx$ 6 GB
REVE fine-tune	10 ep, batch 8	$\approx$ 144 min	$\approx$ 22 GB
LaBraM fine-tune	15 ep, batch 8	$\approx$ 118 min	$\approx$ 12 GB
SAE training (1 layer)	20 ep, 500 K tokens, batch 256	$\approx$ 5–15 min	$\approx$ 4 GB
SD training	60 ep, 500 K tokens	$\approx$ 15–40 min	$\approx$ 6 GB
TCAV	10-fold LR + per-concept probe	$\approx$ 3–8 min	$\approx$ 4 GB
Steering (1 config)	whole dataset, train+val+test	5–15 min	$\approx$ 6 GB

F Public-corpus replication

TUAB. SleepFM is finetuned on the TUAB v3.0.1 normal/abnormal labels with a subject-grouped split (test AUROC 0.907 on 3,576 windows) and analysed with the identical pipeline. Abnormality steering (off-target: age) is selective in 8/8 SAE seeds ($\tilde{\Delta} = 0.150 \pm 0.031$; PCA 0.150, ICA 0.139, random dictionary 0.004, centroid 0.000). TUAB is adult-only, so age is a young (< 40) versus old (≥ 60) contrast; it is weakly encoded (AUROC$_0$ 0.72) and selective in 3/8 seeds ($\tilde{\Delta} = 0.071 \pm 0.043$).

TUEV and [33]. Mean-pooled 10 s (TUEV) and 60 s (TUSL) windows of the TUAB-finetuned SleepFM are classified against background with a logistic probe over 20 subject-disjoint splits (30% test). Table 8 reports the mean over 10 SAE seeds. Steering selectivity for each epileptiform event is the drop of its own probe AUROC minus the mean drop of the other two events, relative to the same quantity under 20 random clamps of equal size.

Table 6: Separable / Entangled / Dead fractions under varied enrichment p thresholds ($E = 8$, $k = 8$).

Encoder (layer)	$p < 0.01$	$p < 0.05$ (default)	$p < 0.1$
SleepFM (L3)	0.47 / 0.28 / 0.25	0.39 / 0.47 / 0.14	0.32 / 0.60 / 0.09
REVE (L9)	0.52 / 0.14 / 0.35	0.51 / 0.30 / 0.19	0.46 / 0.42 / 0.12
LaBraM (L12)	0.09 / 0.01 / 0.89	0.12 / 0.04 / 0.85	0.13 / 0.05 / 0.83

Table 7: **Excess selectivity $\tilde{\Delta}$ by direction source** (SleepFM L3, $E = 1$, $k = 8$; identical clamping, probes, and permutation null). SAE: mean $\pm$ sd over 8 independently seeded SAEs. Other sources are deterministic; $^*p < 0.05$.

Direction source	Abnormality	Age	Med. (psych.)	Med. (ASM)	Sex
SAE (ours)	0.154 ± 0.054	0.217 ± 0.062	0.078 ± 0.030	0.118 ± 0.043	0.043 ± 0.039
PCA	0.193^*	0.077	-0.025	0.204^*	0.103
ICA	0.098^*	0.131^*	0.084^*	0.086^*	0.043^*
Dense CAV	0.017^*	0.035^*	0.036^*	0.056^*	0.059^*
Random dictionary	0.001	0.001	0.011	-0.012	-0.004
Centroid steering	0.000	0.000	0.000	0.000	0.000

G Population-level spectral steering

Abnormality is steered on adult recordings and age within abnormal recordings, so neither contrast is confounded by the other. For abnormality we clamp every feature whose weight in a logistic abnormality probe on SAE activations exceeds 5% of the maximum; for age, every feature enriched for an age group (subject-level Mann–Whitney, BH-adjusted $p < 0.05$). SleepFM uses $E = 8$, $k = 64$; LaBraM and REVE $E = 1$, $k = 8$.

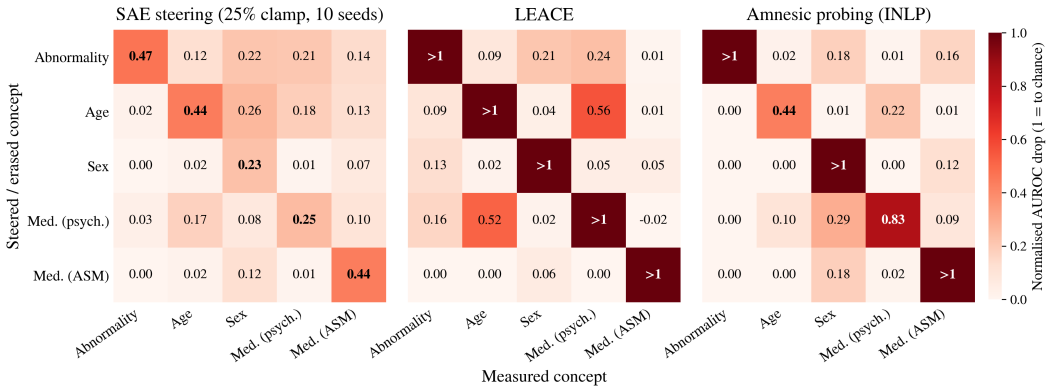


Figure 8: **Steering one concept, measuring all five** (SleepFM L3). Normalised AUROC drop $(\mathrm{AUROC}_{\mathrm{before}} - \mathrm{AUROC}_{\mathrm{after}})/(\mathrm{AUROC}_{\mathrm{before}} - 0.5)$ of each measured concept (columns) after intervening on the row concept: SAE steering (top 25% TCAV-ranked features clamped to the target centroid; mean of 10 SAE seeds), LEACE, and amnesic probing (INLP, 30 iterations). Values > 1 mean the eraser drove the probe below chance. Sex is only weakly encoded (baseline AUROC 0.64), so its normalised entries are noisy.

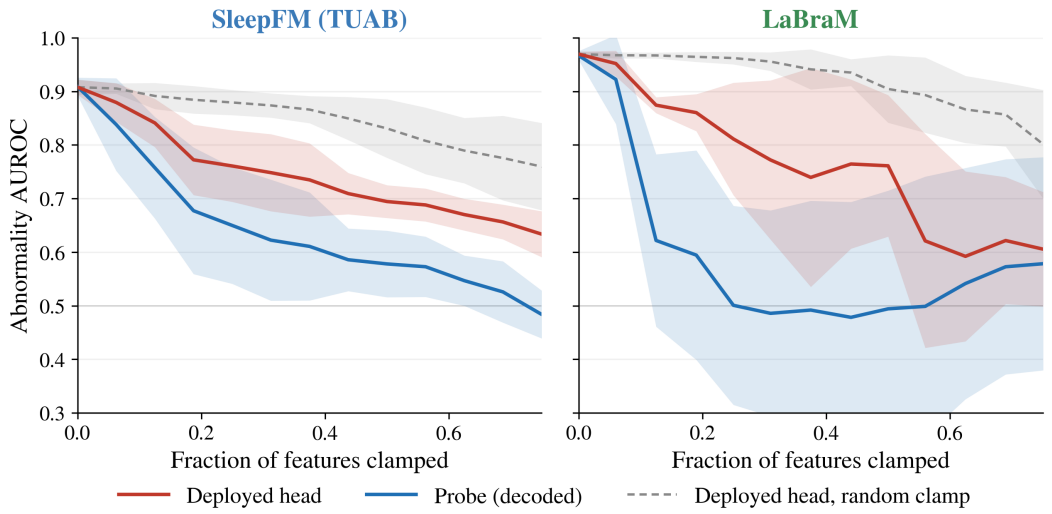


Figure 9: **Interventions propagated through the deployed classifier.** Abnormality AUROC of the finetuned classifier head (red) and of the frozen probe on decoded embeddings (blue) as the fraction of TCAV-ranked features clamped to the normal centroid increases; dashed: the head under random clamping of the same number of features. Mean $\pm$ sd over 10 SAE seeds. SleepFM is the TUAB-finetuned model at its last layer; LaBraM at its last layer. REVE is not shown: its operating layer is intermediate, and we did not propagate the clamped activations through its remaining blocks to the head.

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract and introduction clearly state the four contributions (cross-architecture pipeline, clinical semanticity taxonomy, selectivity metric via concept steering,

Table 8: TUEV / TUSL separability (AUROC vs. background) and TUEV steering selectivity (50% clamp). Mean ± sd over 10 SAE seeds; embedding values do not depend on the SAE seed.

Corpus	Event	n_{pos}	Embedding	SAE features	Steering excess selectivity
TUEV	Spike and sharp wave (SPSW)	74	0.850	0.833 ± 0.015	$+0.067 \pm 0.095$
TUEV	GPED	971	0.973	0.951 ± 0.013	-0.028 ± 0.065
TUEV	PLED	820	0.926	0.907 ± 0.013	$+0.008 \pm 0.051$
TUSL	Slowing	94	0.666	0.623 ± 0.016	–
TUSL	Seizure	92	0.701	0.632 ± 0.037	–

and mechanistic spectral explanations), and the experimental results in Sections 4.1–4.6 directly support each of these claims.

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